SUBMISSION SUMMARY · Q3 2026

Northline repair & maintenance, *reconciled to the cent.*

Finance's Q3 R&M is built from the two ERPs by the eight README rules. It closes against Accounting's figure through two items, each with an accounting reason. The bridge lives in the code: the run fails if anything is left unexplained.

Thiago Farias 2 November 2026 Python + DuckDB · 14 tests · `make all`

01 RESULT

Two numbers, one explained gap.

The 31,929.78 difference is fully explained. Nothing is plugged.

FINANCE Q3 R&M · USD

14,154,722.25

ERP data only, by the eight README rules. Part 1 file: **rm_q3_by_branch.csv**, 13 rows.

ACCOUNTING REPORTED · USD

14,186,652.03

The ledger close total, including a manual accrual and the September FX lookup.

DIFFERENCE	RECONCILING ITEMS	UNEXPLAINED	DATA ISSUES
31,929.78	2	0.00	15
Accounting – Finance	Each with evidence, who is right and what to change	The run fails if this is not zero	Found, proven and treated in the log

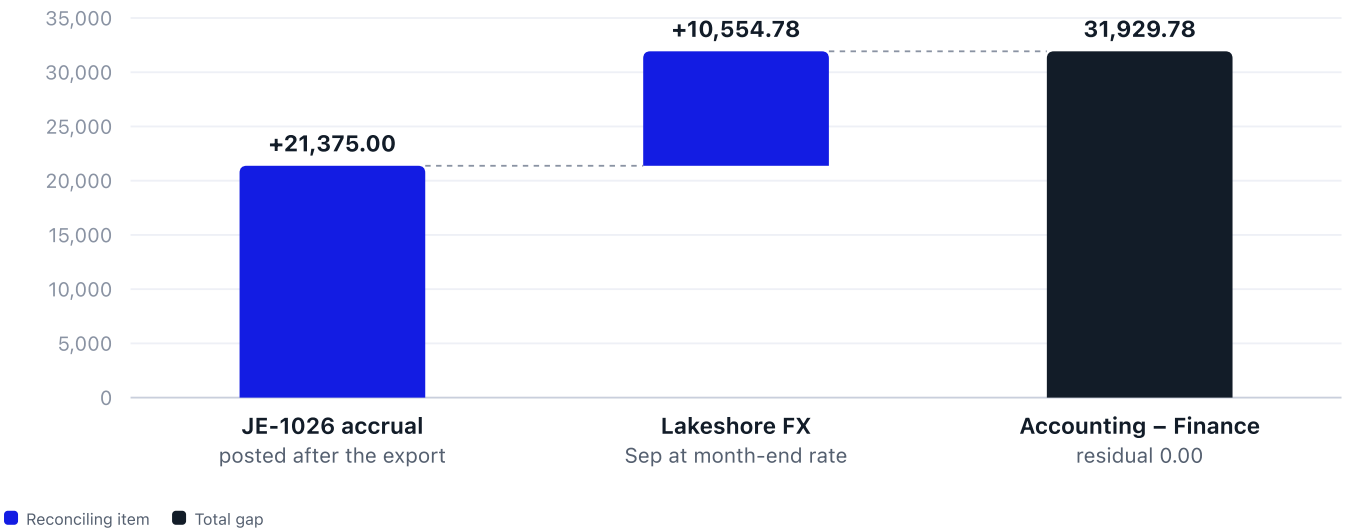
02 PART 3 · BRIDGE

From Finance to Accounting.

Each step is computed from the data, never typed in.

Two items explain the gap; residual 0.00

USD, gap only (Finance = 0)



ITEM	USD	WHO IS RIGHT	WHAT SHOULD CHANGE
Finance Q3 R&M	14,154,722.25		
JE-1026: gearbox overhaul accrual (RIV, 6150), posted 30-Sep 22:14, after the last ERP export at 18:00	21,375.00	BOTH Accounting may accrue a pending invoice; Finance is ERP-only (rule 7)	Reverse the accrual when the invoice posts in ERP-A, or Q4 counts it twice
Lakeshore FX: September at month-end 0.7398 instead of the average 0.7236 · CAD 651,529.93 × 0.0162	10,554.78	FINANCE P&L policy is the monthly average (rule 5)	Fix the close template's FX lookup for the close month
= Accounting reported	**14,186,652.03**		
Residual	**0.00**		

?

How the FX item was found

Testing "Accounting used the month-end rate" for all three months does not close. Dividing the leftover by each month's rate difference does: only September gives exactly Lakeshore's September CAD total. Accounting used month-end for the close month only. Also checked and shown at 0.00 with the reason: five JEs on non-R&M accounts, one JE posted 2-Oct, the 7345-RM-WELD account and the suspected double posts, which are in Accounting's total too.

03 PART 1 · OUTPUT

R&M by branch and month.

USD, in the required `Branch,Month,RM_USD` format. Elmwood reports under Oakdale from 1-Aug (rule 4), so it has July only; Lakeshore is each month's CAD total × that month's average rate, rounded once (rules 5 and 8).

BRANCH	JUL 2026	AUG 2026	SEP 2026	Q3
RIV · Riverton	1,598,866.50	1,701,777.60	1,555,546.51	4,856,190.61
OAK · Oakdale	1,479,513.89	1,434,056.50	1,418,481.69	4,332,052.08
PIN · Pinecrest	1,053,910.47	1,194,058.95	1,095,823.05	3,343,792.47
ELM · Elmwood	324,978.39	—	—	324,978.39
LKS · Lakeshore	417,996.82	408,264.82	471,447.06	1,297,708.70
Total	**4,875,266.07**	**4,738,157.87**	**4,541,298.31**	**14,154,722.25**

04 PART 2 · DATA QUALITY

15 issues, each with its proof.

"Error if untreated" is how far the Q3 total would move if the issue were ignored. The log is generated by the same code as the CSV, so the two cannot disagree.

ID	ISSUE	HOW IT WAS PROVEN	TREATMENT	ROWS	ERROR IF UNTREATED
A1	24-Aug export is a re-run of 22-Aug	780 of 780 keys repeated, identical after cleaning	DEDUPLICATED latest copy kept; stops if copies disagree	780	1,125,583.73
A2	Branch typed 14 ways ("riv", "Riverton ", " OAK")	Distinct values of Whse	NORMALISED alias table; a new spelling stops the run	5,355	6,296,090.47
A3	Account as "6100.0" or " 6100"	Accounts that are not clean 4-digit codes	NORMALISED before rule 1	1,323	1,674,972.31
A4	ISO dates mixed with slash dates	As MM/DD all fall 0–9 days before their export, like ISO dates; as DD/MM only 11 would	PARSED ISO, then MM/DD	545	617,931.42
A5	June and October lines	Date outside 1-Jul to 30-Sep	EXCLUDED rule 3	1,326	1,903,570.78
A6	Capex operations on R&M accounts	Oper 75/100 on 6100/6150/6200	EXCLUDED rule 2	168	448,957.44
A7	Elmwood lines after closure	ELM dated from 1-Aug	REASSIGNED to OAK, rule 4	97	0.00
A8	Customer number on R&M lines	CustNum filled on R&M accounts	KEPT the account decides, rule 1	3,524	0.00
A9	Reversals, 8 crossing a month	All 60 match an original line	KEPT in their own month, rule 3	60	0.00
A10	Suspected double posts	Identical except the Line number; 25 pairs are not chance	FLAGGED kept; tested one-line switch	25	0.00
B1	SUBTOTAL rows inside the ledger	Each checked against its detail (31/31)	EXCLUDED they repeat the detail	31	1,297,708.70
B2	Amount stored as text ("2,992.80")	Excel cell type is text	PARSED a non-number stops the run	18	17,092.54
B3	DD/MM/YYYY dates	First part reaches 31; second never above 10	PARSED day-first	1,473	0.00
B4	7345-RM-WELD missing from the account map	Ledger accounts vs map	INCLUDED rule 6; map to be fixed	23	73,601.96
B5	June and October postings	Date outside Q3	EXCLUDED rule 3	151	0.00

A = ERP-A (US branches, USD). B = ERP-B (Lakeshore, CAD); its error column is converted to USD.

05 JUDGEMENT

Three calls worth discussing.

Where the README did not decide, the data did.

PROOF, NOT ASSUMPTION

The date format

Slash dates read as MM/DD land 0–9 days before the export that carried them, the same lag as ISO dates. As DD/MM, only 11 of 582 would.

OWNER DECIDES

Double posts kept, flagged

They look duplicated, but Accounting's total only closes with them in: they are in the ledger. Removing them is the ERP-A owner's call, so removal is a tested one-line switch.

RULE OVER MAP

7345-RM-WELD included

Rule 6 says every 73xx-RM account is R&M, even though Accounting's map misses it. The bridge confirms Accounting counted it too; only the map needs fixing.

06 HOW IT WORKS

Every line can say why it is in or out.

Python and DuckDB, money in exact decimals. Each README rule is one parameter in `rules_config.py`, one flag per line and one test.

STEP 1

Load as text

Every file as raw text with its row number. Nothing silently coerced.

STEP 2

Type and clean

Dates, accounts, amounts; the raw value kept next to the clean one.

STEP 3

One flag per rule

r1 account, r2 capex, r3 period, r4 branch, r6 73xx-RM, re-export, double post.

STEP 4

Aggregate, then bridge

Sum by branch-month, round last; bridge must reach 0.00.

!

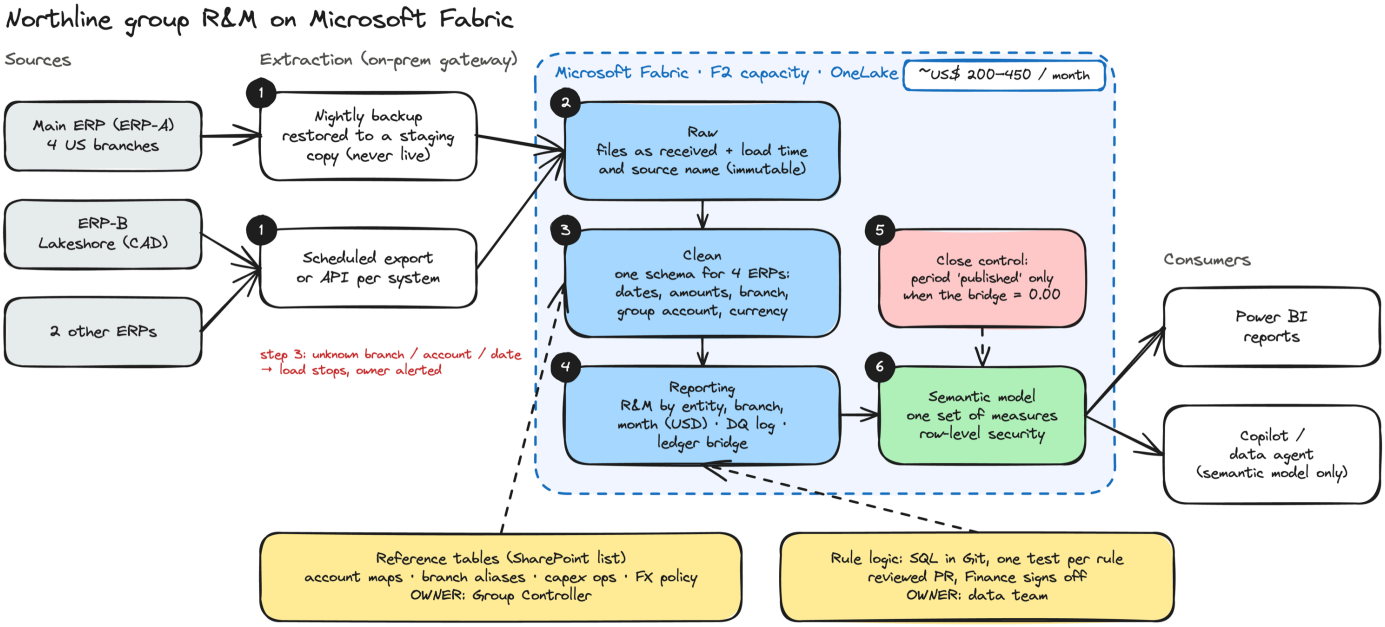
The run stops instead of guessing

An unknown branch spelling, an unparsable date or amount, re-export copies that disagree, a subtotal that does not match its detail, a month without an FX rate, or a bridge that does not close: each stops the run and names the line. Two clean runs produce byte-identical outputs.

07 PART 4 · DESIGN

Microsoft Fabric, sized small.

Northline already runs Microsoft 365 and Power BI, and the volume is under 1 GB a year. The hard part is trust, not scale.



~USD 200–450

per month: F2 capacity reserved, storage, Pro authors (list prices)

0 live queries

on the main ERP: IT restores the nightly backup and the pipeline reads the copy

30 days

from inventory to the October close reconciled to the cent

Who owns what

- **Finance** owns the reference tables: account maps, branch aliases, FX policy, in an approved SharePoint list.
- **Data team** owns the SQL in Git, one test per rule; any rule change is a reviewed PR Finance signs off.

Controls

- A period is published only when the bridge to the ledger closes.
- Copilot answers only from the semantic model, with the same measures as the reports, tested on known questions first.

08 PART 5 · AI

The AI typed the code; the judgement is mine.

Claude Code in VS Code, as a pair programmer with shell access. Everything it produced has a check around it. Its three mistakes, and how each was caught:

MISTAKE 1

A wrong count

29 suspected double posts instead of 25: the flag was joined on a key the re-exported copy shares. Caught against an independent profiling script; now guarded by a test.

MISTAKE 2

A reversed recommendation

It proposed removing the double posts. The bridge showed they are real ledger postings, so they are kept and flagged.

MISTAKE 3

A blind spot

Its bridge search changed the FX rate for all three months at once. Dividing the residual by each month's rate difference found the close-month-only pattern.

09 NEXT STEPS

Open questions and how to run it.

For Accounting

1. Why was September translated at the month-end rate? Likely the close template's FX lookup.
2. Can JE-1026 be reversed when the gearbox invoice posts in ERP-A?
3. Can 7345-RM-WELD be added to the Lakeshore account map?
4. Can the ERP-A owner confirm whether the 25 suspected double posts are real?

Run it

```
make all   # venv + rebuild every output + 14 tests
```

Every rule is a one-line change in `src/rm/rules_config.py`. If a change moves Finance's number away from the ledger, the run stops and says which item the bridge is missing.

IN THE ZIP	CONTENTS
`README.md`	Start here: result, how to run, decisions and assumptions
`output/rm_q3_by_branch.csv`	Part 1, required format
`docs/02_data_quality_log.md`	Part 2, with the line funnel chart
`docs/03_reconciliation.md`	Part 3, every item with evidence
`docs/04_design_memo.pdf`	Part 4, one page
`docs/05_ai_log.md`	Part 5
`src/rm/` · `tests/`	Python + DuckDB pipeline and the 14 tests
`data/raw/`	The data pack exactly as received

Northline Industrial Services is fictional; the data is synthetic, as stated in the brief.